TOPIC 3
TRAVERSE

找 Latit / Depart = Rec
找 Bearing / Jarak = Pol

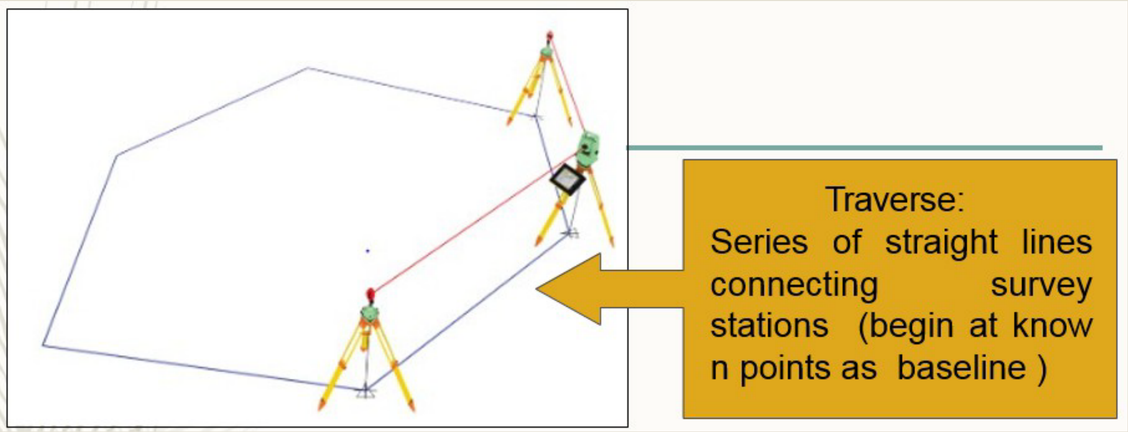
OBJECTIVE

Student should be able to:

1) Understand latitude and departure
2) Calculate linear misclose and traverse accuracy
3) Calculate traverse adjustment
4) Calculate missing data with non-adjacent reference and adjacent reference.
5) Calculate Traverse Area using DMD Methos
6) Calculate the coordinate computation

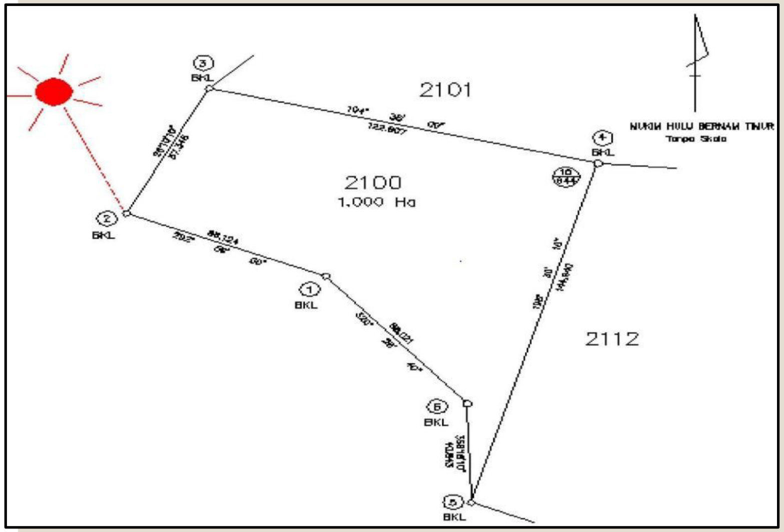
TRAVERSE

DEFINITION



- Traversing is that type of survey in which a number of connected survey lines form the frame work.
- There are **TWO (2)** types of the traverse:
  1. CLOSE TRAVERSE
  2. OPEN TRAVERSE

CLOSE TRAVERSE



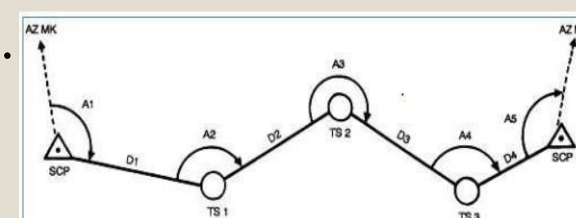
- A closed traverse starts at a point and ends at the same point or at a point whose relative position is known.
- 2 Types of close traverse:

   i. **Loop Traverse**

   - ➢ A loop traverse starts and ends on a station of assumed coordinates

   ii. **Connective Traverse**

   - ➢ It begins and ends at points or lines of known position.

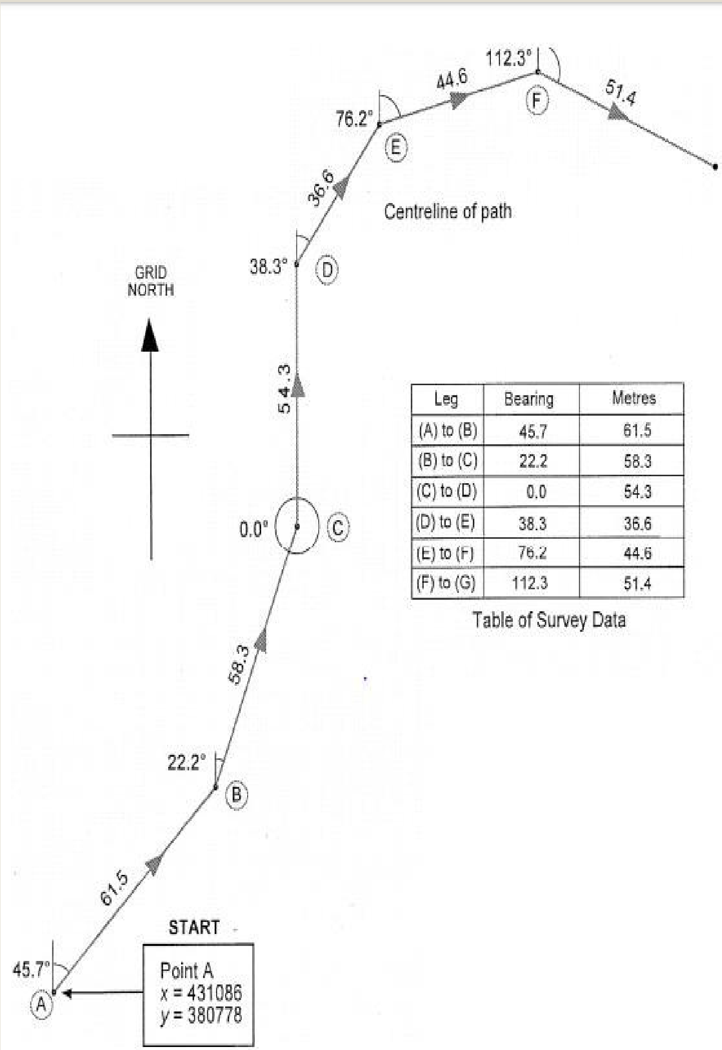


CRM Station

BKL/BKB Station

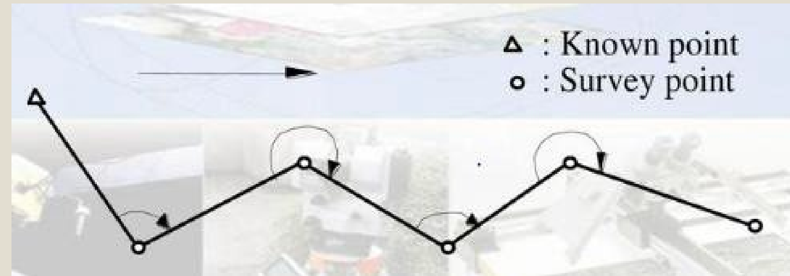
Coordinate Station

OPEN TRAVERSE



Leg	Bearing	Metres
(A) to (B)	45.7	61.5
(B) to (C)	22.2	58.3
(C) to (D)	0.0	54.3
(D) to (E)	38.3	36.6
(E) to (F)	76.2	44.6
(F) to (G)	112.3	51.4

Table of Survey Data



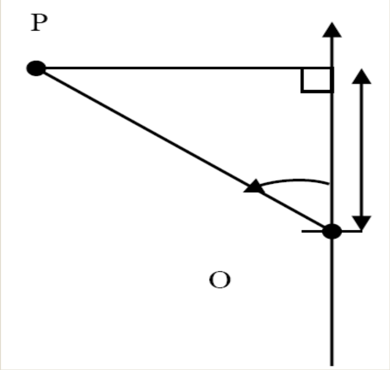
- An open traverse originates at a starting station, proceeds to its destination and ends at a station whose relative position is not previously known.

LATITUDE AND **DEPARTURE**

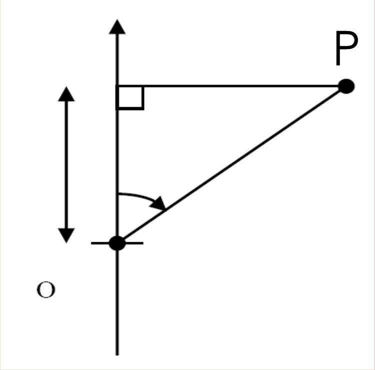
(Kiraan Latit & Dipat)

LATITUDE

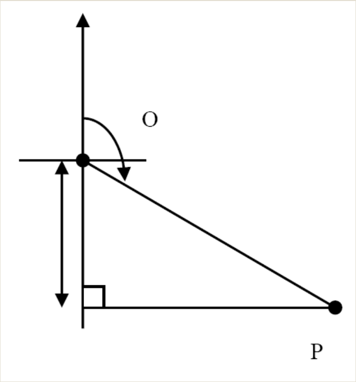
- Projections on the reference meridian (Y axis – NORTH & SOUTH)



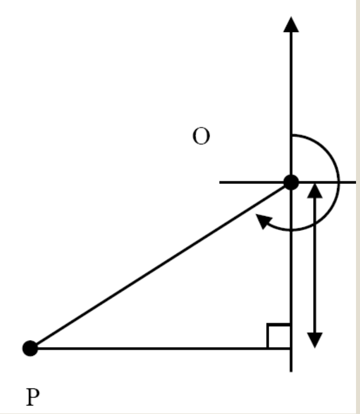
Quadrant 4 = Latit (+ve)
= North (N)



Quadrant 1 = Latit (+ve)
= North (N)



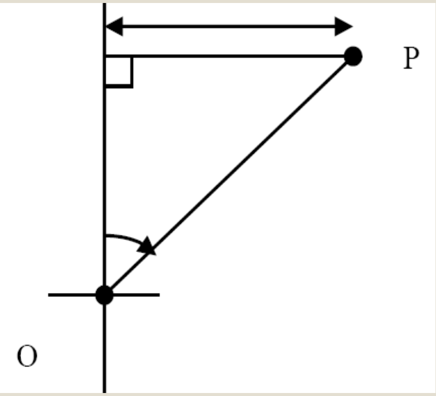
Quadrant 2 = Latit (-ve)
= South (S)



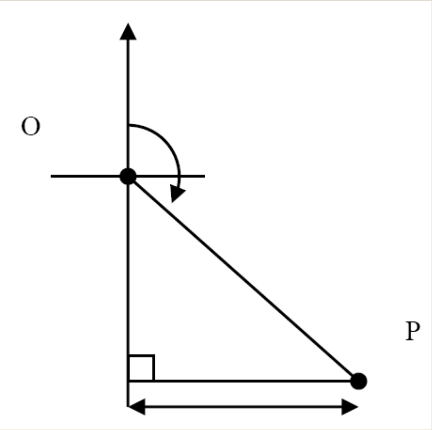
Quadrant 3 = Latit (-ve)
= South (S)

DEPARTURE

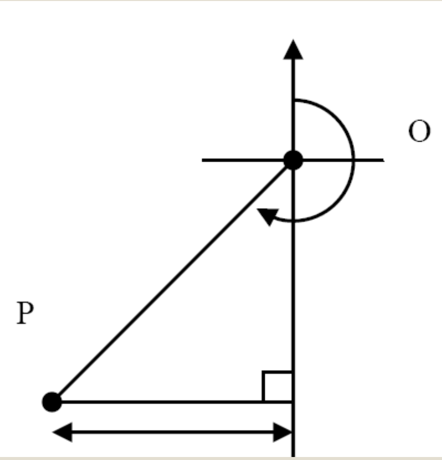
- Projections on the right angle to the reference meridian. (X axis – EAST & WEST)



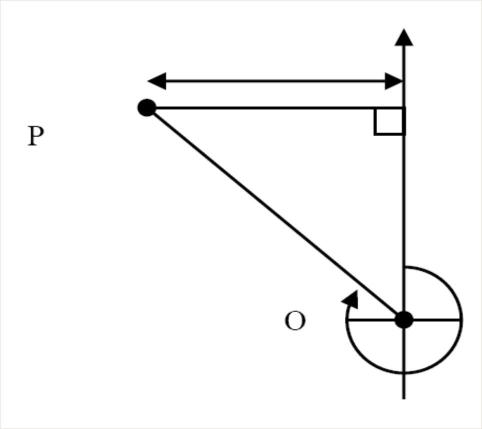
Quadrant 1 = Depart (+ve)
= East (E)



Quadrant 2 = Depart (+ve)
= East (E)



Quadrant 3 = Depart (-ve)
= West (W)



Quadrant 4 = Depart (-ve)
= West (W)

COMPUTATION OF LATITUDE AND DEPARTURE

FORMULA:

Latitude : Distance x Cos $\theta°$
Depart : Distance x Sin $\theta°$

OR

Using Calculator:

Shift → **-** Rec(Distance , Bearing) → **=**
Get Latitude Value

RCL → **tan**
Get Departure Value

EXAMPLE:

Compute Latitude and departure for line:

LINE	BEARING	DISTANCE
1 - 2	30° 00′ 00″	60.123 m
1 - 3	100° 30′ 00″	80.326 m

Line 1-2:

Latit = Distance x Cos $\theta°$
= 60.123 x Cos 30° 00′ 00″
= 52.068

Dipat = Distance x Sin $\theta°$
= 60.123 x Sin 30° 00′ 00″
= 30.062

Line 1-3:

Latit = Distance x Cos $\theta°$
= 80.326 x Cos 100° 30′ 00″
= -14.638

Dipat = Distance x Sin $\theta°$
= 80.326 x Cos 100° 30′ 00″
= 78.981

EXERCISE

1. Calculate Latit and depart for line:

Garisan	Bering	Jarak (m)	Latit	Dipart
1-2	45° 00′ 00″	65.008		
2-3	135° 00′ 00″	85.001		
3-4	225° 00′ 00″	65.002		
4-1	315° 00′ 00″	85.005		

2. Calculate Latit and depart for line:

Garisan	Bering sukuan	Jarak (m)	Latit	Dipart
1-2	U 40° 50′ 00″ T	500.000		
2-3	S 35° 40′ 00″ T	400.000		
3-4	S 30° 50′ 00″ B	300.000		

Exercises

Data-data dalam jadual 5.1 adalah hasil cerapan bering bagi sebuah trabas tertutup.

Hitungkan latit dan dipat setiap garisan tersebut.

Stesen	Bering	Jarak (m)
1		
2	07° 39′ 10″	57.363
3	98° 30′ 20″	52.770
4	99° 01′ 50″	30.486
5	169° 36′ 40″	60.159
6	279° 54′ 40″	34.263
1	277° 39′ 10″	67.643

Jadual 5.1

Tentukan nilai latit dan dipat bagi setiap garisan trabas seperti yang ditunjukkan dalam jadual 5.4.

Stesen	Bering	Jarak (m)
1		
2	$31^\circ\ 41'\ 10''$	47.170
3	$345^\circ\ 45'\ 00''$	48.710
4	$294^\circ\ 54'\ 30''$	67.390
5	$222^\circ\ 37'\ 10''$	67.840
6	$167^\circ\ 10'\ 40''$	59.220
1	$95^\circ\ 35'\ 30''$	81.560

Jadual 5.4

<u>LINEAR MISCLOSE</u>

(Kiraan Tikaian Lurus)

Calculate Linear Misclose And Traverse Accuracy

Linear Traverse (Linear Misclose) = A calculated quantity as an indicator of the quality level achieved for any working of close Traverse.

Formula Linear Misclose (LM):

$$\boxed{LM = \frac{\sqrt{\Delta L^2 + \Delta D^2}}{\text{Total Distance}}}$$

EXAMPLE

Garisan Dari-Ke	Bering	Jarak	Latit		Dipat	
			U	S	T	B
1						
2	63° 30′ 00″	63.264	28.229		56.617	
3	77° 25′ 00″	75.119	16.365		73.315	
4	173° 43′ 30″	82.147		81.655	8.979	
5	231° 55′ 00″	87.273		53.831		68.694
1	322° 19′ 00″	114.829	90.876			70.195
	Jumlah	422.632	135.470	135.486	138.911	138.889
			ΔL = 0.016		ΔD = 0.022	

$$LM = \frac{\sqrt{\Delta L^2 + \Delta D^2}}{\text{Total Distance}}$$

$$= \frac{\sqrt{0.016^2 + 0.022^2}}{422.632}$$

$$= 0.000064365$$

$$= 1 \; / \; 0.000064365$$

= **15536.26131 #**

EXERCISE

Based on data below, calculate:

i. Latitude and departure for each line

ii. Linear Misclose for close traverse

Garisan Dari-Ke	Bering	Jarak
1		
2	51° 20′ 00″	137.045
3	180° 00′ 00″	190.466
4	238° 05′ 30″	17.944
1	321° 15′ 00″	146.560

DMD

①找错误
- 改
- U +ve
  S -ve
  T -ve
  B +ve
- 2 method
  • $\frac{\text{Latit} \times 错误值}{\Sigma \text{ Latit}}$
  • $\frac{\text{Jarak} \times 错误值}{\Sigma \text{ Jarak}}$

② 2x Latit / 2x Depart
i) 自己
ii) 自己+上面+上面

③ 2x Latit (Depart) / 2x Depart (Latit)
- 两个算到一样

④ Area
$2 \times \text{Area} = \Sigma (2 \times \text{Latit}) \times \text{Depart}$

$\text{Area} = \frac{\Sigma (2 \times \text{Latit}) \times \text{Depart}}{2}$

TRAVERSE ADJUSTMENT

(Pelarasan Trabas)

TRAVERSE ADJUSTMENT

TWO (2) methods of traverse adjustment:

1. **Transit method**

$$\frac{\text{Latit} \times \text{错误值}}{\text{总数}}$$

Formula:

$$\text{Pembetulan Latit}_{1\text{-}2} = \frac{\text{Latit}_{1\text{-}2} \times \text{Perbezaan Latit}}{\text{Jumlah Latit}}$$

$$\text{Pembetulan Dipat}_{1\text{-}2} = \frac{\text{Dipat}_{1\text{-}2} \times \text{Perbezaan Dipat}}{\text{Jumlah Dipat}}$$

2. **Bowditch method**

Formula:

$$\text{Pembetulan Latit}_{1\text{-}2} = \frac{\text{Garisan}_{1\text{-}2} \times \text{Perbezaan Latit}}{\text{Jumlah Jarak}}$$

$$\text{Pembetulan Dipat}_{1\text{-}2} = \frac{\text{Garisan}_{1\text{-}2} \times \text{Perbezaan Dipat}}{\text{Jumlah Jarak}}$$

Adj Line 1-2 = Latit 1-2 x

TRANSIT METHOD

EXAMPLE

Compute correction of the latitudes and departures using Transit Method.

Garisan Dari-Ke	Bering	Jarak	Latit		Dipat	
			U	S	T	B
1						
2	169° 36′ 40″	60.170		59.184	10.850	
3	095° 02′ 40″	8.298		0.730	8.266	
1	342° 18′ 40″	62.879	59.906			19.106

Answer:

Garisan Dari-Ke	Bering	Jarak	Latit		Dipat	
			U	S	T	B
1						
2	169° 36′ 40″	60.170		59.184	10.850	
3	095° 02′ 40″	8.298		0.730	8.266	
1	342° 18′ 40″	62.879	59.906			19.106
	Jumlah		59.906	59.914	19.116	19.106
			$\Delta L = 0.008$		$\Delta D = 0.010$	
	Jumlah latit utara dan selatan = 119.820					
	Jumlah dipat timur dan barat = 38.222					

错误的相差

Steps:

1. Calculate Latit & Dipat
2. Get ΔL & ΔD value
3. Get total value for latit & dipat

4. Using formula for Transit Method to get adjustment for each line.

$$\text{Adj Line}_{1\text{-}2} = \frac{\text{Latit}_{1\text{-}2} \times \Delta L}{\sum \text{Latit}}$$

$$= \frac{59.184 \times 0.008}{119.820}$$

$$= 0.004$$

$$\text{Adj Line}_{1\text{-}2} = \frac{\text{Dipat}_{1\text{-}2} \times \Delta D}{\sum \text{Latit}}$$

$$= \frac{10.850 \times 0.010}{38.222}$$

$$= 0.003$$

$$\text{Adj Line}_{2\text{-}3} = \frac{\text{Latit}_{2\text{-}3} \times \Delta L}{\sum \text{Latit}}$$

$$= \frac{0.730 \times 0.008}{119.820}$$

$$= 0.000$$

$$\text{Adj Line}_{2\text{-}3} = \frac{\text{Dipat}_{2\text{-}3} \times \Delta D}{\sum \text{Latit}}$$

$$= \frac{8.266 \times 0.010}{38.222}$$

$$= 0.002$$

$$\text{Adj Line}_{3\text{-}1} = \frac{\text{Latit}_{3\text{-}1} \times \Delta L}{\sum \text{Latit}}$$

$$= \frac{59.906 \times 0.008}{119.820}$$

$$= 0.004$$

$$\text{Adj Line}_{3\text{-}1} = \frac{\text{Dipat}_{3\text{-}1} \times \Delta D}{\sum \text{Latit}}$$

$$= \frac{19.106 \times 0.010}{38.222}$$

$$= 0.005$$

5. Final Result

+/- 错误值

Garisan Dari-Ke	Bering	Jarak	Latit		Dipat	
			U (+)	S (-)	T (-)	B (+)
1						
2	169° 36′ 40″	60.170		59.184 - 0.004	10.850 - 0.003	
3	095° 02′ 40″	8.298		0.730 - 0.000	8.266 - 0.002	
1	342° 18′ 40″	62.879	59.906 + 0.004			19.106 + 0.005
	Jumlah		59.906	59.914	19.116	19.106
			$\Delta L = 0.008$		$\Delta D = 0.010$	
	Jumlah latit utara dan selatan = 119.820					
	Jumlah dipat timur dan barat = 38.222					

Note: *Pelarasan untuk* ***semua latit utara*** *adalah +ve dan pelarasan untuk semua* ***latit selatan*** *adalah –ve. Ini kerana jumlah latit selatan adalah lebih besar berbanding dengan jumlah latit utara.*

Note: *Pelarasan untuk* ***semua dipat timur*** *adalah –ve dan pelarasan untuk* ***semua dipat barat*** *adalah +ve. Ini kerana jumlah dipat timur lebih besar berbanding dengan dipat barat.*

Garisan Dari-Ke	Bering	Jarak	Latit		Dipat	
			U	S	T	B
1						
2	169° 36′ 40″	60.170		59.180	10.847	
3	095° 02′ 40″	8.298		0.730	8.264	
1	342° 18′ 40″	62.879	59.910			19.111
	Jumlah		59.910	59.910	19.111	19.111
			$\Delta L = 0.000$		$\Delta D = 0.000$	

EXERCISE

Based on data given, Compute correction of the latitudes and departures using Transit Method.

Garisan Dari-Ke	Bering	Jarak
1		
2	110° 15′ 20″	60.123
3	200° 30′ 40″	40.567
4	271° 15′ 40″	42.234
5	355° 15′ 20″	43.256
1	013° 43′ 40″	15.217

BOWDITCH METHOD

EXAMPLE

Compute correction of the latitudes and departures using Bowditch Method.

Garisan Dari-Ke	Bering	Jarak	Latit		Dipat	
			U	S	T	B
1						
2	275° 02′ 00″	8.299	0.728			8.267
3	279° 54′ 10″	34.265	5.893			33.754
4	277° 39′ 40″	67.643	9.018			67.039
1	098° 09′ 50″	110.170		15.645	109.054	

Answer:

Garisan Dari-Ke	Bering	Jarak	Latit		Dipat	
			U	S	T	B
1						
2	275° 02′ 00″	8.299	0.728			8.267
3	279° 54′ 10″	34.265	5.893			33.754
4	277° 39′ 40″	67.643	9.018			67.039
1	098° 09′ 50″	110.170		15.645	109.054	
	Jumlah	220.377	15.639	15.645	109.054	109.060
			$\Delta L = 0.006$		$\Delta D = 0.006$	

Steps:

1. Calculate Latit & Dipat
2. Get ΔL & ΔD value
3. Get total value for distance

4. Using formula for Bowditch Method to get adjustment for each line.

$$\text{Adj Line}_{1\text{-}2} = \frac{\text{Distance}_{1\text{-}2} \times \Delta L}{\sum \text{Distance}}$$

$$= \frac{8.299 \times 0.006}{220.377}$$

$$= 0.000$$

$$\text{Adj Line}_{1\text{-}2} = \frac{\text{Distance}_{1\text{-}2} \times \Delta D}{\sum \text{Distance}}$$

$$= \frac{8.299 \times 0.006}{220.377}$$

$$= 0.000$$

$$\text{Adj Line}_{2\text{-}3} = \frac{\text{Distance}_{2\text{-}3} \times \Delta L}{\sum \text{Distance}}$$

$$= \frac{34.265 \times 0.006}{220.377}$$

$$= 0.001$$

$$\text{Adj Line}_{2\text{-}3} = \frac{\text{Distance}_{2\text{-}3} \times \Delta D}{\sum \text{Distance}}$$

$$= \frac{34.265 \times 0.006}{220.377}$$

$$= 0.001$$

$$\text{Adj Line}_{3\text{-}4} = \frac{\text{Distance}_{3\text{-}4} \times \Delta L}{\sum \text{Distance}}$$

$$= \frac{67.643 \times 0.006}{220.377}$$

$$= 0.002$$

$$\text{Adj Line}_{3\text{-}4} = \frac{\text{Distance}_{3\text{-}4} \times \Delta D}{\sum \text{Distance}}$$

$$= \frac{67.643 \times 0.006}{220.377}$$

$$= 0.002$$

$$\text{Adj Line}_{4\text{-}1} = \frac{\text{Distance}_{4\text{-}1} \times \Delta L}{\sum \text{Distance}}$$

$$= \frac{110.170 \times 0.006}{220.377}$$

$$= 0.003$$

$$\text{Adj Line}_{4\text{-}1} = \frac{\text{Distance}_{4\text{-}1} \times \Delta D}{\sum \text{Distance}}$$

$$= \frac{110.170 \times 0.006}{220.377}$$

$$= 0.003$$

5. Final Result

Garisan Dari-Ke	Bering	Jarak	Latit U	Latit S	Dipat T	Dipat B
1						
2	275° 02′ 00″	8.299	0.728 + 0.000			8.267 - 0.000
3	279° 54′ 10″	34.265	5.893 + 0.001			33.754 - 0.001
4	277° 39′ 40″	67.643	9.018 + 0.002			67.039 - 0.002
1	098° 09′ 50″	110.170		15.645 - 0.003	109.054 + 0.003	
	Jumlah	220.377	15.639	15.645	109.054	109.060
			ΔL = 0.006		ΔD = 0.006	

Note: *Pelarasan untuk* ***semua latit utara*** *adalah* ***+ve*** *dan pelarasan untuk* ***semua latit selatan*** *adalah* ***– ve****. Ini kerana jumlah latit selatan lebih besar daripada jumlah latit utara.*

Note: *Pelarasan untuk* ***semua dipat timur adalah +ve*** *dan pelarasan untuk* ***semua dipat barat adalah –ve.*** *Ini kerana jumlah dipat barat lebih besar berbanding jumlah dipat timur.*

Garisan Dari-Ke	Bering	Jarak	Latit U	Latit S	Dipat T	Dipat B
1						
2	275° 02′ 00″	8.299	0.728			8.267
3	279° 54′ 10″	34.265	5.894			33.753
4	277° 39′ 40″	67.643	9.020			67.037
1	098° 09′ 50″	110.170		15.642	109.057	
	Jumlah	220.377	15.642	15.642	109.057	109.057
			ΔL = 0.000		ΔD = 0.000	

EXERCISE

Based on data given, Compute correction of the latitudes and departures using Bowditch Method.

Garisan Dari-Ke	Bering	Jarak
1		
2	300° 12′ 00″	36.392
3	044° 24′ 30″	42.792
4	114° 37′ 00″	37.706
5	180° 00′ 30″	31.334
1	266° 48′ 30″	32.772

TRAVERSE AREA CALCULATION USING DOUBLE MERIDIAN DISTANCE (DMD) METHOD

(Pengiraan Keluasan Trabas dengan menggunakan kaedah DMD)

Calculate area of traverse using Double Meridian Method (DMD).

Garisan Dari-Ke	Latit		Dipat	
	U	S	T	B
1				
2		60.113	60.109	
3		45.963		45.965
4	60.108			60.110
1	45.968		45.966	
Jumlah	106.076	106.076	106.075	106.075
	Beza latit (ΔL) = 0.000		Beza dipat (ΔD) = 0.000	

Answer:

Line From To	Latit		Depart		2x Latit	2x Dipat
	N (+)	S (-)	E (+)	W (-)		
1						
2		60.113	60.109	(+)	-60.113	60.109
3		45.963	(+) [=]	45.965	- 166.189	74.253
4	60.108			60.110	-152.044	-31.822
1	45.968		45.966		-45.968	-45.966
Jumlah	106.076	106.076	106.075	106.075		
	ΔL = 0.00		ΔL = 0.00			

1

Line From To	Latit		Depart		2x Latit	2x Dipat
	N (+)	S (-)	E (+)	W (-)		
1						
2		60.113	60.109	(+)	-60.113	60.109
3		45.963	(+) [=]	45.965	-166.189	74.253
4	60.108			60.110	-152.044	-31.822
1	45.968		45.966		45.968	-45.966
Jumlah	106.076	106.076	106.075	106.075		
	ΔL = 0.00		ΔL = 0.00			

*①自己
②自己+上面+上面
③ "
④ "
*注意看+ve/-ve

Method:

Latit

2x Latit (1-2) = Latit (1-2)
= - 60.113

2x Latit (2-3) = 2x Latit (1-2) + Latit (1-2) + Latit (2-3)
= - 60.113 + (-60.113) + (-45.963)
= -166.189

2x Latit (3-4) = 2x Latit (2-3) + Latit (2-3) + Latit (3-4)
= -166.189 + (-45.963) + (60.108)
= -152.044

2x Latit (4-5) = 2x Latit (3-4) + Latit (3-4) + Latit (4-1)
= -152.044 + (60.108) + (45.968)
= 45.968

Dipat

2x Dipat (1-2) = Dipat (1-2)
= 60.109

2x Dipat (2-3) = 2x Dipat (1-2) + Dipat (1-2) + Dipat (2-3)
= 60.109 + (60.109) + (-45.965)
= 74.253

2x Dipat (3-4) = 2x Dipat (2-3) + Dipat (2-3) + Dipat (3-4)
= 74.253 + (-45.965) + (-60.110)
= -31.822

2x Dipat (4-5) = 2x Dipat (3-4) + Dipat (3-4) + Dipat (4-1)
= -31.822 + (-60.110) + (45.966)
= -45.966

Note: *For final checking, value of 2x Latit (4-1) must be same with Latit (4-1)*
For final checking, value of 2x Dipat (4-1) must be same with Dipat (4-1)

2

Line From To	Latit		Depart		2x Latit	2x Dipat	(2x Latit) x Dipat	(2x Dipat) x Latit
	N (+)	S (-)	E (+)	W (-)				
1								
2		60.113	60.109		-60.113	60.109	- 3613.332	- 3613.332
3		45.963		45.965	- 166.189	74.253	7638.877	- 3412.891
4	60.108			60.110	-152.044	-31.822	9139.365	- 1912.757
1	45.968		45.966		- 45.968	-45.966	- 2112.965	- 2112.965
Jumlah	106.076	106.076	106.075	106.075				

To calculate the area of traverse, choose only 1: **(2x Latit) x Dipat** **OR** **(2x Dipat) x Latit**

(2x Latit) x Dipat

[(2x Latit) x Dipat] (1-2) = (-60.113) x (60.109)
= - 3613.332

[(2x Latit) x Dipat] (2-3) = (-166.189) x (-45.965)
= 7638.877

[(2x Latit) x Dipat] (3-4) = (-152.044) x (-60.110)
= 9139.365

[(2x Latit) x Dipat] (4-1) = (-45.968) x (45.966)
= - 2112.965

(2x Dipat) x Latit

[(2x Dipat) x Latit] (1-2) = (60.109) x (-60.113)
= - 3613.332

[(2x Dipat) x Latit] (2-3) = (74.253) x (-45.963)
= - 3412.891

[(2x Dipat) x Latit] (3-4) = (-31.822) x (60.108)
= - 1912.757

[(2x Dipat) x Latit] (4-1) = (- 45.966) x (45.968)
= - 2112.965

3

Line From To	(2x Latit) x Dipat	(2x Dipat) x Latit
1		
2	- 3613.332	- 3613.332
3	7638.877	- 3412.891
4	9139.365	- 1912.757
1	- 2112.965	- 2112.965
Jumlah	11051.945	-11051.945

To calculate the area of traverse, choose only 1:
(2x Latit) x Dipat **OR**
(2x Dipat) x Latit

Formula using DMD Method:

2 x Area = ∑ (2x Latit) x Dipat

Area = 11051.945 ÷ 2
= **5750.973 m²**

OR

2 x Area = ∑ (2x Dipat) x Latit

Area = -11051.945 ÷ 2
= **-5750.973 m²**

EXERCISE

Calculate area of traverse using Double Meridian Method (DMD).

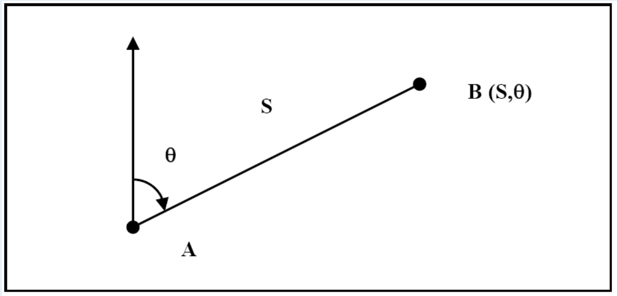
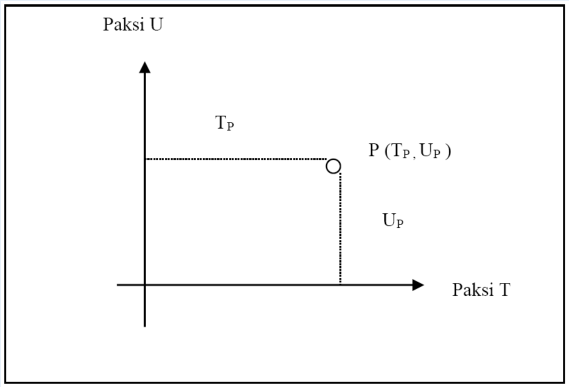
Garisan Dari-Ke	Latit		Dipat	
	U	S	T	B
1				
2		112.321		71.556
3		7.355		121.118
4	97.811			54.119
5	105.514		55.612	
6		46.939	112.809	
1		36.710	78.372	

Answer:

Area = 31843.429 m²

COORDINATE

Point on a plane surface can be determined by two systems: Polar coordinates and Cartesian coordinates.

POLAR COORDINATE	CARTESIAN COORDINATE
Involves angle, bearing and distance	• Used in survey work • Using 2 axes 90° to each other • 4 axis (N,S,E & W)
 θ = Angle S = Bearing A = Distance	 Position of each station traverse relating to all other points is expressed in the form of E and N coordinates for Station P.

CASE 1: Coordinate Of A Survey Station With A Given Bearing, Distance and Coordinate Of Other Station

Example

Bearing 1-2 = 75° 15’ 00”

Distance 1-2 = 15.750 m

Coordinate Stn 1 = N 50, E 50

Calculate coordinate of Station 2.

Using Calculator:

Shift → - Rec(Distance , Bearing) → =

Get Latitude Value

RCL → tan

Get Departure Value

N 50
E 50

Koordinat utara stesen 2	= Koordinat utara stesen 1 + latit 1-2
Koordinat timur stesen 2	= Koordinat timur stesen 1+ dipat 1-2
Koordinat utara stesen 3	= Koordinat utara stesen 2 + latit 2-3
Koordinat timur stesen 3	= Koordinat timur stesen 2 + dipat 2-3

Begitulah kiraan koordinat untuk stesen-stesen ukur yang seterusnya.

ANSWER:

1 Find Latit & Dipat for line 1-2

L = 4.100

D = 15.231

2 Find coordinate for Stn 2

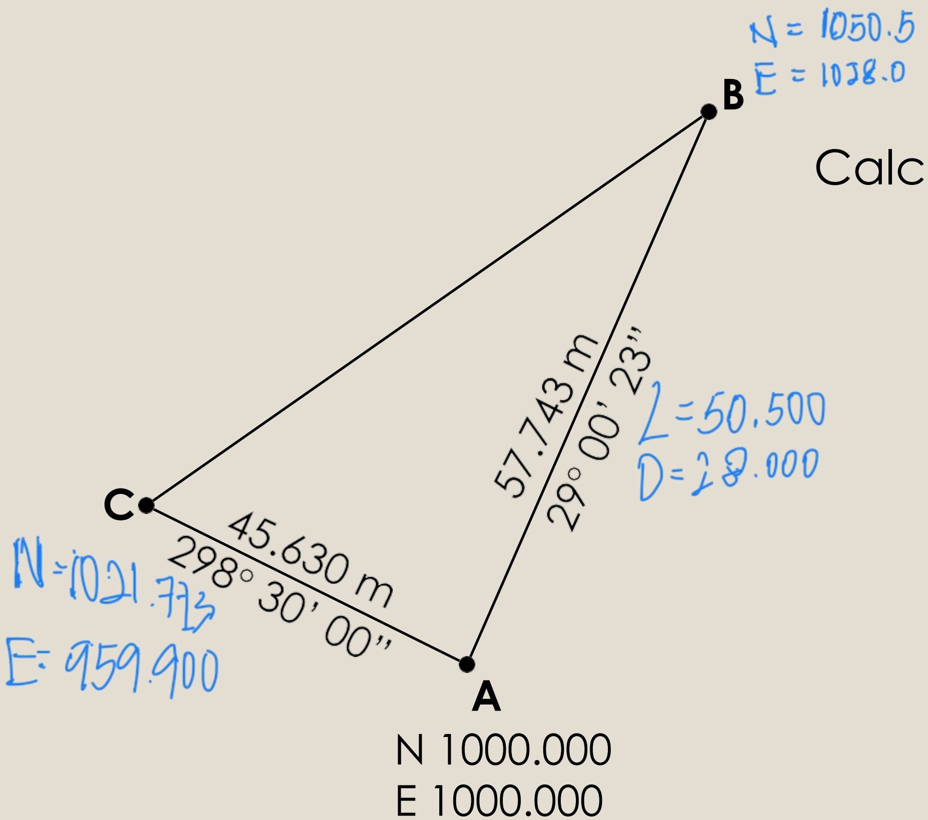
N = N50 + 4.100

= N 54.100

E = E50 + 15.231

= E 65.231

EXERCISE



Calculate coordinate station B & C.

ANSWER:

STN B : N 1050.500
E 1028.000

STN C : N1021.773
E 959.900

CASE 2: Compute bearing and distance from two known coordinates

Using Calculator:

Shift → [-] Pol(Latit , Dipat) → [=]
Get Distance Value

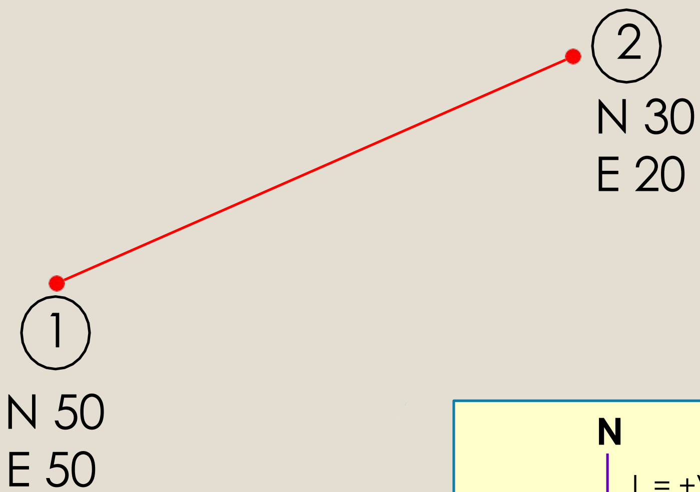
[RCL] → [tan]
Get $\theta°$ (Bearing) Value

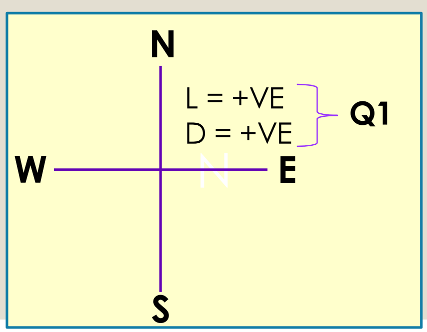
Example

Coordinate Stn 1 = N 10, E 10

Coordinate Stn 2 = N 30, E 20

Calculate bearing and distance of Station 1-2.





ANSWER:

1 Find Latit & Dipat for line 1-2

L = N 50 – N 30
= +20

D = N 50 – N 20
= +30

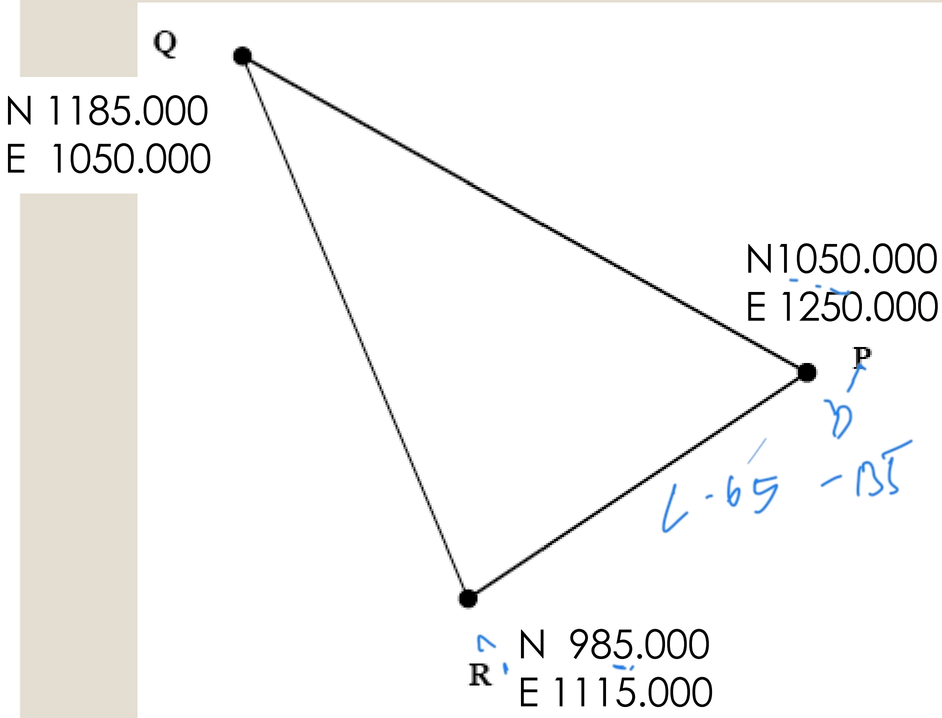
2 Using Calculator, find bearing and distance for line 1-2

Distance 1-2 = 36.056 m

$\theta°$ = 56° 18' 36"

So, Bearing 1-2 = 56° 18' 36"

EXERCISE



Calculate bearing and Distance PR, RQ & QP

ANSWER:

Bearing PR : 244° 17’ 24”
Distance PR : 149.833 m

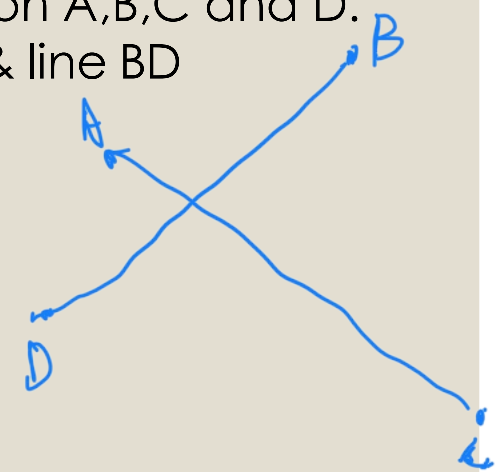
Bearing RQ : 341° 59’ 45”
Distance RQ : 210.297 m

Bearing QP : 124° 01’ 10”
Distance QP : 241.299 m

CASE 3: Calculate coordinate intersection between two lines

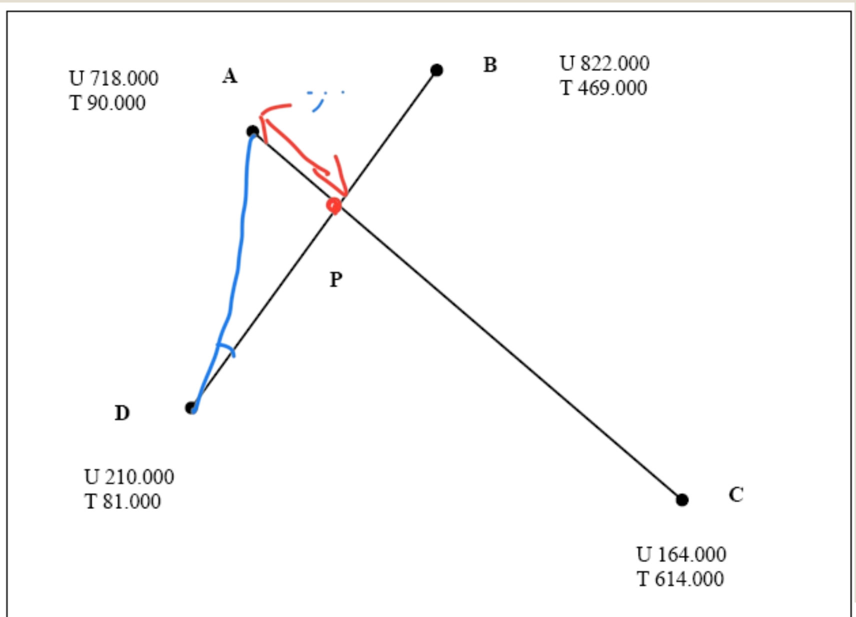
Stesen Dari-Ke	Koordinat	
	Utara	Timur
A	718.000	90.000
B	822.000	469.000
C	164.000	614.000
D	210.000	81.000

Question: Based on data given, sketch the position of station A,B,C and D. Then, calculate coordinate intersection between line AC & line BD

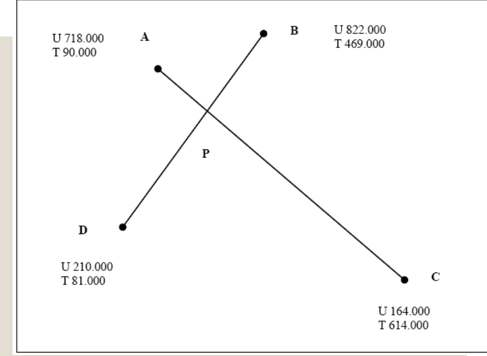


ANSWER:

1. Sketch the position of 4 points.



2 Find Latit & Dipat for line AC



L = N 718.000 – N164.000
= + 554

D = E90.000 – E614.000
= - 551

Using Calculator, find bearing for line AC

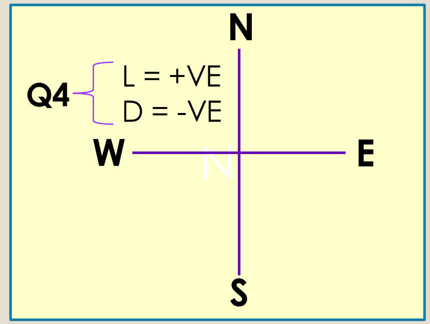
$\theta° = 43° 24' 21''$

So,

Bearing AC = $360° - \theta°$

= $360° - 43° 24' 21''$

= 316° 35′ 39″



Using Calculator:

Shift → **-** Pol(Latit , Dipat) → **=**

Get Distance Value

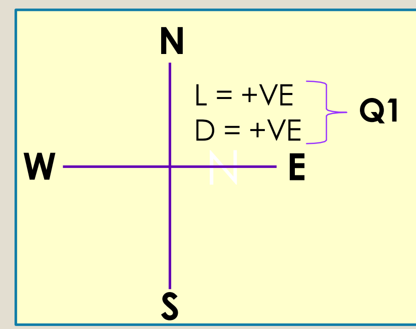
RCL → **tan**

Get θ° (Bearing) Value

3 Find Latit & Dipat for line BD

L = N 822.000 – N210.000
= + 612

D = E469.000 – E81.000
= + 388



Using Calculator, find bearing for line BD

$\theta° = 32° 22' 27''$

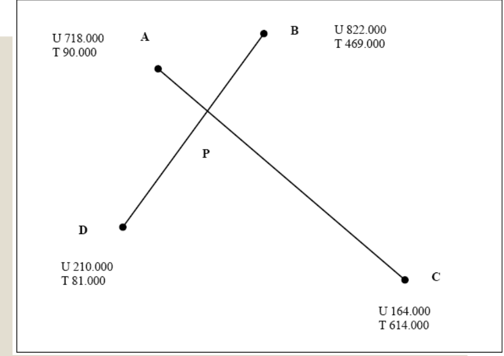
So,

Bearing DB = $32° 22' 27''$ *(Bila tolak nilai koordinat dari B ke D, bering yang dapat adalah dari D ke B)*

Bearing BD = $180° + 32° 22' 27''$

= 212° 22' 27"

4 Find bearing and distance AD



L = N 210.000 – N718.000
= - 508

D = E81.000 – E90.000
= - 9

Using Calculator, find bearing & distance AD

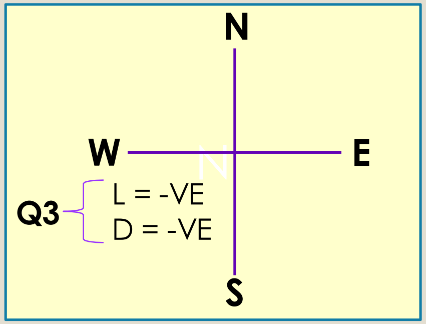
Distance AD = 508.080 m

$\theta° = 1° \, 00' \, 54''$

Bearing AD = $180° + \theta°$

$= 180° + 1° \, 00' \, 54''$

$= 181° \, 00' \, 54''$



Using Calculator:

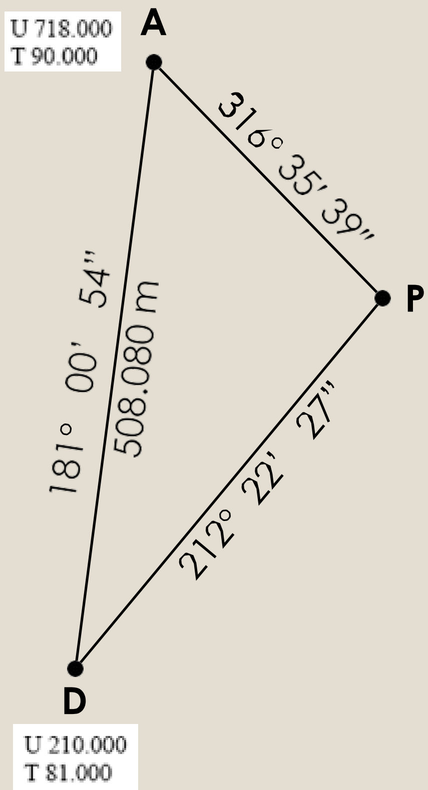
Shift → **-** Pol(Latit , Dipat) → **=**

Get Distance Value

RCL → **tan**

Get θ° (Bearing) Value

5 Find Distance AP



i. Find ∠D

$\angle D$ = Bearing DP – Bearing DA

$= (212^\circ\ 22'\ 27'' - 180^\circ) - (181^\circ\ 00'\ 54'' - 180^\circ)$

$= 32^\circ\ 22'\ 27'' - 1^\circ\ 00'\ 54''$

$= 31^\circ\ 21'\ 33''$

ii. Find ∠P

$\angle P$ = Bearing PA – Bearing PD

$= 316^\circ\ 35'\ 39'' - 212^\circ\ 22'27''$

$= 104^\circ\ 13'\ 12''$

iii. Find distance AP

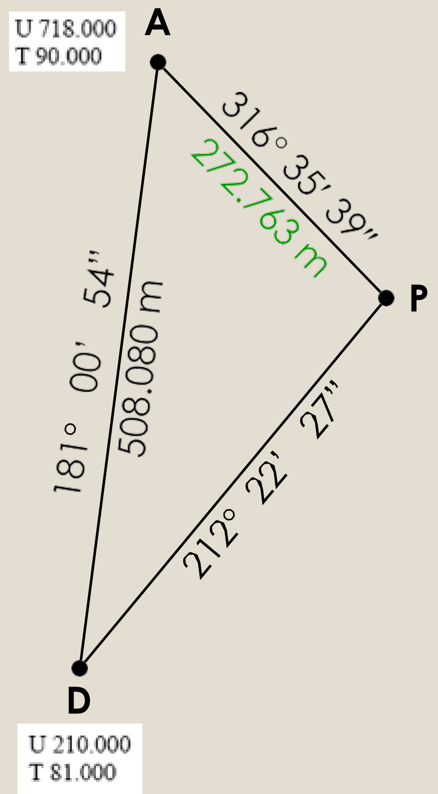
$$\frac{AP}{\text{Sin } D} = \frac{AD}{\text{Sin } P}$$

$$\frac{AP}{\text{Sin } 31^\circ\ 21'\ 33''} = \frac{508.080}{\text{Sin } 104^\circ\ 13'\ 12''}$$

$$AP = \frac{508.080 \times \text{Sin } 31^\circ\ 21'\ 33''}{\text{Sin } 104^\circ\ 13'\ 12''}$$

$= 272.763$ m

6 Find Coordinate Station P



i. Find Latit & Dipat for line AP

(Arah bering yang diambil mestilah daraipada A ke P)

L = - 198.164

D = 187.432

198.1636054

-187.4322269

Using Calculator:

Shift → - Rec(Distance , Bearing) → =

Get Latitude Value

RCL → tan

Get Departure Value

ii. Find coordinate for Station P

N = N718.000 + (-198.164)

= **N 916.164**

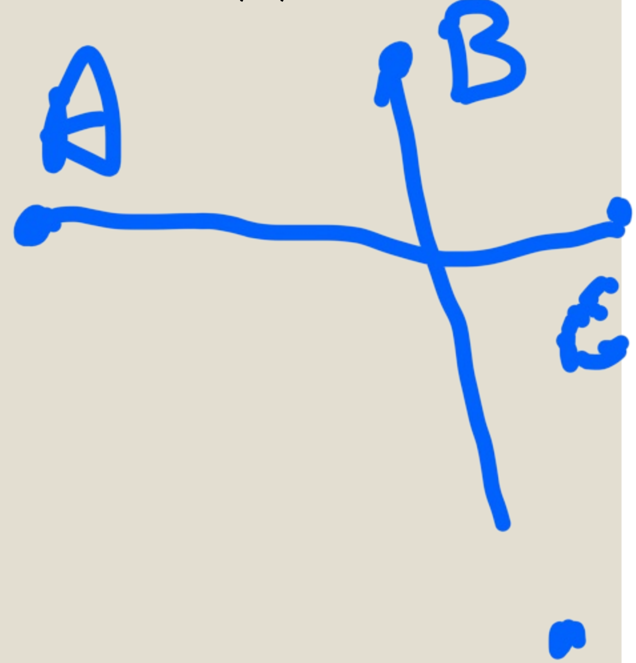
E = E90.000 + 187.432

= **E 272.432**

EXERCISE

Stesen Dari-Ke	Koordinat	
	Utara	Timur
A	1100.000	1250.000
B	1170.700	1320.700
C	1170.700	1520.700
D	736.880	1625.450

Question: Based on data given, sketch the position of station A,B,C and D. Then, calculate coordinate intersection line X.



ANSWER:

Coordinate X :
N 1126.563
E 1351.705

CASE 4: Calculate coordinate and area of a close traverse

Example

Calculate coordinate for stn 2,3,4 & 5

Line From - To	Bearing	Distance	Latit		Dipat		2x Latit	2x Dipat	Koordinat	
			N +	S -	E +	W -			N / S	E/W
1									500.000	500.000
2				20.818	56.405					
3				37.997		14.214				
4			0.928			42.224				
5			43.106			3.578				
1			14.781		3.611					

Coordinate 2 = N 500 - 20.818
= 479.182

= E 500 + 56.405
= 556.405

Coordinate 3 = N 479.182 - 37.997
= 441.185

= E 556.405 - 14.214
= 542.191

CASE 4: Calculate coordinate and area of a close traverse

Example

Line From - To	Bearing	Distance	Latit		Dipat		2x Latit	2x Dipat	Koordinat	
			N	S	E	W			N / S	E/W
1									500.000	500.000
2				20.818	56.405					
3				37.997		14.214				
4			0.928			42.224				
5			43.106			3.578				
1			14.781		3.611					

Coordinate 4 = N 441.185 + 0.928
= 442.113

= E 542.191 – 42.224
= 499.967

Coordinate 5 = N 442.113 + 43.106
= 485.219

= E 499.967 – 3.578
= 496.389

Coordinate 1 = N 485.219 + 14.781
= 500.000

= E 496.389 + 3.611
= 500.000

Line From - To	Bearing	Distance	Latit		Dipat		2x Latit	2x Dipat	Koordinat	
			N	S	E	W			N / S	E/W
1									**500.000**	**500.000**
2				20.818	56.405				479.182	556.405
3				37.997		14.214			441.185	542.191
4			0.928			42.224			442.113	499.967
5			43.106			3.578			485.219	496.389
1			14.781		3.611				500.000	500.000

<u>EXERCISE</u>

Garisan Dari Ke	Bering	Jarak	Latit		Dipat		Dua Kali Latit	Dua Kali Dipat	Koordinat			
			U	S	T	B				Selatan		Timur
1												
2			45.968		45.966							
3				60.113	60.109					**100.000**		**200.000**
4				45.963		45.965						
1			60.108			60.110						

Question: Based on data given, calculate coordinate for station 1,2 and 4

<u>ANSWER:</u>

Garisan Dari Ke	Bering	Jarak	Latit		Dipat		Dua Kali Latit	Dua Kali Dipat	Koordinat			
			U	S	T	B				Selatan		Timur
1										85.855		93.925
2			45.968		45.966					39.887		139.891
3				60.113	60.109					**100.000**		**200.000**
4				45.963		45.965				145.963		154.035
1			60.108			60.110				85.855		93.925

Koordinat	
N / S	E/W
500.000	500.000
479.182	556.405
441.185	542.191
442.113	499.967
485.219	496.389
500.000	500.000

Calculate the area :

Stesen	Koordinat			
Dari Ke		Utara		Barat
1		1070.357		1104.010
2		1027.321		1075.034
3		1018.202		1040.191
4		992.820		1030.334
5		900.000		1100.000
6		926.077		1137.068
7		958.251		1175.113
8		1029.166		1141.274
1		1070.357		1104.010

Calculate the area of stn 1 to 8